

SAHIL GOYAL

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Scholar



GitHub



LinkedIn



Medium

EDUCATION

Indian Institute of Technology Roorkee

2019 - 2024

B.S.+M.S. in Applied Mathematics

Department Rank 1 (Gold Medal)

Overall GPA: **9.44**/10

TECHNICAL SKILLS

Languages and Core Frameworks: Python, C++, JAX (Flax, Optax, Orbx, Chex), PyTorch, XLA

PUBLICATIONS

- ELT: Elastic Looped Transformers for Visual Generation.** [CVPR-W '26]
Sahil Goyal*, Swayam Agrawal*, Prateek Jain, Sujoy Paul, Aditya Kusupati
- MaGNeTS: Masked Generative Nested Transformers with Decode Time Scaling.** [Paper] [Poster]
Sahil Goyal*, Debapriya Tula*, Gagan Jain, Pradeep Shenoy, Prateek Jain, Sujoy Paul [ICML '25]
- Design-o-meter: Towards Evaluating and Refining Graphic Designs.** [Paper] [Poster] [Survey]
Sahil Goyal*, Abhinav Mahajan*, Swasti Mishra, Prateeksha Udhayan, [WACV '25]
Tripti Shukla, K J Joseph, Balaji Vasanth Srinivasan.
- Emotionally Enhanced Talking Face Generation.** [Paper] [Poster] [Code]
Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Yi Yu, [ICCV-W '23]
Yifang Yin, Rajiv Ratn Shah. [ACM MM-W '23]
- Emotional talking faces: Making videos more expressive and realistic.** [Paper] [Demo]
Sahil Goyal, Shagun Uppal, Sarthak Bhagat, Dhroov Goel, Sakshat Mali, [ACM MM Asia '22]
Yi Yu, Yifang Yin, Rajiv Ratn Shah. [Best Demo paper]

EXPERIENCE

- Pre-Doctoral Researcher at Google DeepMind** June 2024 - Present
- Research in **Efficient ML** (training-time compute efficiency, inference-time compute efficiency, and parameter efficiency) including contributions to **Veo** and **Gemini**, and publications at ICML [2] and CVPR-W [1].
- Research Intern at Adobe Research** May 2023 - July 2023
- Unified framework [3] for automated design evaluation and refinement, enabling iterative improvements for creators.
- Research Intern at MIDAS Lab @IITD** Sep 2021 - Jan 2023
- Multimodal talking face generation network [4, 5], winning **Best Demo Paper** award at ACM MM Asia.
- Machine Learning Intern at PANDO** March 2021 - April 2021
- Developed a **20x** faster 3-D space optimization algorithm for transport logistics.

KEY RESEARCH PROJECTS

- Compression of Pre-trained LLMs** May 2025 - Present
Advisors: *Dr. Sujoy Paul, Dr. Prateek Jain, Dr. Pradeep Shenoy*
- Optimizing **model compression** pipeline for **Gemini**, aiming for **minimal extra training cost**.
 - Building and refining effective **structured-pruning** and **layer-wise distillation** techniques.
 - Initial results show significant quality gains over iso-params and iso-FLOPs baselines.
- Adaptive Compute for Visual Generation** June 2024 - Present
Advisors: *Dr. Sujoy Paul, Dr. Prateek Jain, Dr. Aditya Kusupati*
- Developed an **adaptive computation** method [2] for multi-step generative models using **nested transformers**.

- Achieved **~3x** reduction in inference FLOPs for image/video generation.
- Implemented similar efficient techniques for Google's SOTA video generation model (**Veo**).
- Extending the method for **on-device** generation using **elastic looped transformers**.

Automated Evaluation and Refinement of Graphic Designs

May 2023 - July 2023

Advisors: **Dr. Joseph KJ**, **Dr. Balaji Vasan Srinivasan**

- Built compact **Siamese** networks (**0.5M** params) to score **Graphic Designs**.
- Significantly outperformed GPT-4o and LLaVA-NeXT (7B) in ranking graphic designs.
- Tailored advanced **Genetic Algorithms** for SOTA **Layout refinement** (mean-IOU of **0.54** on Crello dataset).

Emotionally Enhanced Talking Face Generation

Sep 2021 - Jan 2023

Advisors: **Dr. Rajiv Ratn Shah**

- Developed the first **talking face generation** network [4, 5] that incorporated emotions.
- SOTA results on **CREMA-D** dataset (**83.2%** emotion classification accuracy and **FID** metric: **5.29**).
- Deployed the entire system as an interactive web interface.

SELECTED ACHIEVEMENTS

- | | |
|--|------|
| • 2 Patents filed at US Patent Office on Efficient Visual Generation. | 2025 |
| • Among 15 , from 18,000+ applicants, selected for Google's Pre-Doctoral Researcher Program. | 2024 |
| • Department Gold Medal (IIT Roorkee - Applied Mathematics). | 2024 |
| • Best Demo Paper at ACM MM ASIA conference. | 2022 |
| • Gold Medal at Inter IIT Tech Meet. | 2022 |
| • Open Source: PyTorch-Vision (debugged dataloaders, ported tests and updated docs). | 2022 |
| • Awarded ACM SigMM student travel grant for ACM MM Asia. | 2022 |
| • JEE Advanced Rank of 1583 among 1.5M students. | 2019 |
| • 600+ stars on  GitHub . | |

PROJECTS B.C. (BEFORE CHATGPT)

Model Extraction Attacks for Video Classification **[Code]**

Feb 2022 - March 2022

- Successfully extracted models across grey-box (5% original data) and black-box (synthetic data only) settings.
- Developed novel data augmentation, synthetic data generation and knowledge distillation techniques.
- Our method won **gold medal** at Inter IIT Tech Meet.

Global Wheat Detection Challenge 2021 **[Code]**

June 2021 - July 2021

- Implemented and benchmarked SOTA **object detection** models with diverse backbones for wheat-head detection.
- Improved accuracy to **56.2%** using training tweaks (data augmentation, pseudo labeling) and post-processing techniques (weighted boxes fusion, non-maximum suppression).

Multimodal Image-to-Image Translation **[Code]** **[Blog]**

Dec 2020 - Jan 2021

- Implemented **BicycleGAN** from scratch in PyTorch for **edges-to-images** and **denoising images** tasks.
- Reproduced the **results** for paper's diversity-based **LPIPS** benchmark.

VisualML **[Code]** **[Demo]**

May 2020 - July 2020

- Developed a set of interactive, **browser-based ML visualization** demos.
- Engineered in-browser execution by integrating TensorFlow.js, Plotly.js, and numeric libraries.

POSITIONS OF RESPONSIBILITY

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|--|----------------|
| • Reviewer at NeurIPS, ICML, ICLR, CVPR, ECCV, ICCV, AAMAS, WACV conferences. | 2024 - Present |
| • Mentor at Data Science Group IITR | 2020 - 2024 |
| • Undergraduate Teaching Assistant at Academic Reinforcement Program IITR | 2021 |